

SOLUTION ACCELERATOR · MULTI-JURISDICTION COMPLIANCE

Compliance Pack Accelerator

A Databricks-native multi-jurisdiction compliance platform with pluggable regulation packs — DPDP India 2023, UK GDPR, EU GDPR, CCPA, PIPEDA, and beyond. Discover personal data, capture granular consent, fulfill data-subject requests, score compliance gaps, and generate audit-ready DPIAs from a single Unity Catalog spine. Compliance applies per data subject: each principal's jurisdiction routes to the pack governing them, so an Indian SaaS with UK clients applies DPDP rules to Indian principals and UK GDPR rules to UK principals, in the same database, without manual sorting.

Get started

What's in scope

The compliance problem in one paragraph

Reduce multi-regulator penalty exposure and prove compliance to auditors on demand

Modern data businesses face a growing matrix of data-protection regulators. India's DPDP Act 2023 imposes penalties up to **₹250 crore**; UK GDPR up to **£17.5 million or 4% of global turnover**; EU GDPR up to **€20 million or 4%**; CCPA, PIPEDA, LGPD, APPI each their own ceilings. A multinational data business runs all of these against the same database. Most organizations cannot answer the basic question — *where is our personal data, which regulation governs each row, who has consent for what, and can we erase it on request?* — without weeks of manual spreadsheet work across siloed systems. Existing tools cover slices (PII scanning, consent capture, DSR portals) but rarely connect them, and almost never on the same governance plane the data lives on.

This accelerator delivers an end-to-end multi-jurisdiction compliance platform on the Databricks Data Intelligence Platform: PII discovery, consent intelligence, data-subject rights fulfillment, penalty-weighted gap analysis, retention enforcement, persona-scoped Genie agents, and AI-assisted DPIA generation — every layer sharing the same Unity Catalog governance spine, every rule routed by the principal's jurisdiction column to the regulation pack governing them. The seed deployment is DPDP India 2023; UK GDPR is the next pack; EU GDPR / CCPA / PIPEDA / LGPD / APPI drop in as 9-yaml-file authoring exercises with no core code change.

Who it's for

FOR COMPLIANCE & PRIVACY

Living personal data register

Auto-classified PII inventory across every ingestion pattern. Penalty-weighted gap analysis tied to the active regulation pack. DPIA generation as a one-click Genie call.

FOR DATA ENGINEERING

Three ingestion patterns, one governance layer

Auto Loader, Lakeflow Connect, and Lakehouse Federation all feed the same classifier and same UC controls. Adding a new source is *one row* in `bronze.data_sources` — no classifier code changes.

FOR LEGAL & GENERAL COUNSEL

DSR-ready, audit-ready, multi-language

Discovery + erasure scripts produce signed audit bundles. Notice versions in 10 Indian languages, with machine-translation watermarks. Persona-scoped Genie agents enforce who-sees-what at query time.

What's included

PII Discovery & Classification

DISCOVERY

Sixteen-pattern PII library (eleven universal + five India-specific: Aadhaar, PAN, IFSC, India phone, India passport). Vectorized Lakeflow Declarative Pipelines classifier scans every silver column on every run; UC column tags and masks auto-applied. Non-admin users see redacted values regardless of query path — dashboard, Genie, SQL editor, or REST API.

Consent Intelligence Engine

CONSENT

Append-only consent event log with granular per-purpose tracking, notice-version supersession, and a `has_active_consent()` UDF as the single source of truth. A marketing-eligibility view filters at query time; Delta Change Data Feed on silver tables drives downstream re-aggregation the moment a principal withdraws — no batch lag.

Data Subject Rights Fulfillment

DSR

UC-lineage-aware discovery script identifies every row about a principal across the data estate; the erasure script applies Delta DELETE + VACUUM with policy-respecting preserve lists, signed audit bundles, and a written record into `compliance.dsr_requests`. End-to-end chain is exercised by an automated test that runs in seconds.

Penalty-Weighted Gap Analysis

COMPLIANCE

Nine DPDP rules drive a generated `compliance_gaps` table — one row per (rule × PII column) — with severity tiers, regulation citations, and remediation guidance. Gap counts roll up into a ₹-denominated exposure view (₹250 cr / 150 / 50 / 5 ceilings) so the CFO Genie can surface the loss-of-day question in plain English.

Multi-Source Native Ingestion

INGESTION

Three Databricks-native ingestion paths share one governance plane: Auto Loader (file-arrival landing zones), Lakeflow Connect (managed SaaS connectors like Salesforce), and Lakehouse Federation (foreign catalogs over external systems). Adding a new source is one row in **bronze.data_sources** — the classifier picks it up automatically, no code changes.

Persona-Scoped Governance

GOVERNANCE

Four personas (Chief Compliance Officer, General Counsel, Chief Marketing Officer, Chief Financial Officer) each get a sliced dashboard, a domain-scoped Genie agent with code-driven instructions, and UC-enforced data boundaries. CMO cannot read patient PII even if they write the SQL by hand — UC denies at query time. One-command deploy provisions all four with their grants, ACLs, and Genie spaces.

AI Agents for Privacy Operations

AI AGENTS

DPIA generator and Compliance Q&A run on Foundation Model serving with MLflow tracing, exponential-backoff retries, and content-hashed versioned prompts. The unstructured-data classifier uses **ai_classify** and **ai_extract** for free-text fields regex cannot handle. A headless health-check validates endpoint + experiment + prompts before every deployment.

Multi-Regulation Pack Framework

EXTENSIBILITY

Regulation-agnostic core; everything DPDP-specific lives in **regulations/dpdp_2023/** as nine yaml files (rules, rights, notices, retention, residency, breach SLA, languages, India PII patterns). Adding a UK GDPR or CCPA jurisdiction is authoring nine new yaml files against the same contract — no core changes. The active regime is selected by a single **REGULATION_PACK** environment variable.

By the numbers

36

PII FINDINGS

10

SILVER OBJECTS SCANNED

3

INGESTION PATTERNS

33

UC COLUMN MASKS

21

MULTI-PACK RULES · 462
GAPS

1,000

CONSENT EVENTS · 292
PRINCIPALS

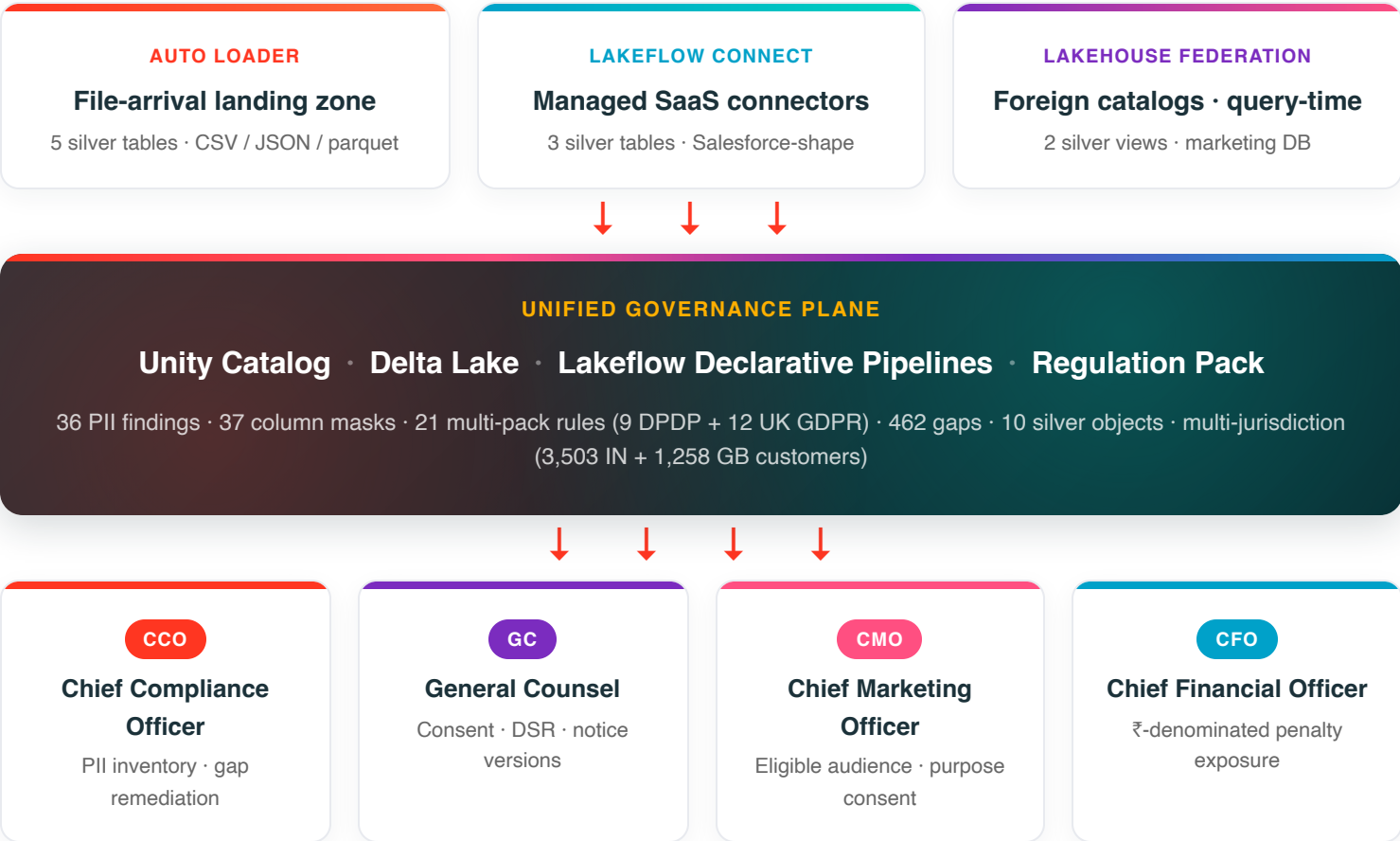
10

NOTICE LANGUAGES

4

PERSONA AGENTS

Architecture at a glance



LAYER	DATABRICKS PRODUCT	WHAT IT DOES
Ingestion	Auto Loader · Lakeflow Connect · Lakehouse Federation	Three patterns, all UC-governed identically
Storage	Delta Lake on Unity Catalog	Bronze + Silver + Gold with CDF for withdrawal propagation
Pipeline	Lakeflow Declarative Pipelines (DLT)	Medallion + classification as DLT tables
Governance	UC column masks · row filters · grants · tags	Per-persona enforcement at query time
Compliance OLTP	Delta (Lakebase on paid tier)	Append-only consent log; DSR request register
AI	Foundation Model APIs · MLflow · Agent Bricks	DPIA + Compliance Q&A + ai_classify
Consumption	AI/BI Dashboards · Genie spaces	Sliced per persona, NL queries with UC enforcement
Deployment	Databricks Asset Bundles	One-command deploy via <code>scripts/deploy_all.sh</code>

Get started

Three commands on a fresh Databricks workspace:

```
# 1. Point the bundle at your workspace (one-time, idempotent)
scripts/configure_workspace_host.sh https://dbc-XXXX.cloud.databricks.com

# 2. Run the full 16-step deploy (UC bootstrap + bundle + synthetic + medallion + SF +
#    federation + bootstrap + tags + masks + filters + multilang + agents + smoke)
scripts/deploy_all.sh

# 3. Wire up the four persona dashboards + Genie spaces
python3 scripts/setup_all_personas.py
```

Verify any time:

```
# Three independent suites; each prints a per-check summary
python3 tests/test_post_deploy_smoke.py      # 10 deploy-gate checks
python3 tests/test_dsr_e2e.py                # 11 DSR-chain assertions
python3 scripts/setup_agent_bricks.py --smoke # 4 Agent Bricks checks
```

Resources

Architecture diagram

[architecture.html](#) — layer-by-layer view of the platform with persona enforcement

Step-by-step test suite

[how_to_test.html](#) — Tests 1–16 with SQL + Genie companions

Modular framework writeup

[modular_framework.html](#) — how the regulation-pack abstraction supports adding UK GDPR / CCPA / others

Solution scope & decisions

[poc_scope.md](#) — capability inventory and design decisions

Persona governance writeup

[persona_governance.md](#) — 2-layer enforcement model and migration path

Business case & ROI

[business_pitch.html](#) — penalty exposure framing, security & ops efficiency benefits

Ready to run it?

Full deploy in ~30 minutes against any Databricks workspace. Idempotent every step, synthetic data only — no real PII anywhere.

[Open the README](#)